What is a Class D Amplifier (Switch Mode Amplifier)

This paper covers the basic functionality of how a class D audio amplifier works. First, the amp compares the input audio signal with a test signal. After the device compares the two inputs, the switches that control the speaker are turned on and off. The speed the switches function at can replicate the input audio signal. The output the switches create is not perfect and requires smoothing out; smoothing out of the output is called filtering.

First, the test signals the audio signal compares to are made by a smaller device called an oscillator. Comparing the two input signals can be accomplished by a device called a comparator. Whenever the audio signal is greater, the output of the comparator will give an on state. When the audio signal is less than the test signal, the output will give an off state. Switches can then be controlled using the on and off states of the comparator.

Switches are made using transistors that can be either on or off when power is supplied to them. In this context, when the transistors are on, it allows the speaker to produce sound. When the transistor is off, the speaker does not produce sound. This switching produces sharp edges in the sound quality. This will cause the speaker to sound distorted and changes the input audio signal. The switches allow the amplifier to be efficient at the cost of introducing this annoying noise to the speaker.

With filtering the annoying noise can be reduced through some well thought out components. Just like a coffee filter, filtering allows certain things to pass through and stops others from going through. Filtering works for this amp because the signal that the switches make has hidden the audio signal. Having a filter is required so that the audio signal can pass through, but the annoying noise will be blocked. After all the filtering the audio signal will be exactly the same as the input audio signal but will instead have a higher power rating.

This audio amplifier is better in efficiency than other traditional audio amplifiers. Switches being used instead of the traditional audio amplifier components leads to a more efficient device. This is also helped by the filtering device in the audio amplifier. This amp can also utilize a control loop, this loop allows the designer to be able to compare the output audio to the input audio and ensure proper operation. If the loop detects any errors the designer can then adjust the switches of the circuit to increase efficiency and sound quality.

In conclusion class D audio amplifiers are power efficient amps that use a combination of comparing, switching and filtering to produce audio. The comparing creates a simple, straightforward, and intuitive way to control the switches in the circuit. The switches allow the entire system to be efficient as they only consume minimal power. Lastly, the filter finds the hidden audio signal in the switches and allows the amplified audio input to come through the speakers.